

MC125: Functions of Single Variable and ODEs

Course Placement: Functions of Single Variable and ODEs is a core course for first year students of BTech MnC program.

Course Format: It is **3 hours' lecture and 1-hour tutorial** every week.

Course Contents:

UNIT-1: Real numbers system: Properties of real numbers, sequence and series. Limit, continuity and differentiability of functions of single variables.

UNIT-2: Differentiation: Implicit differentiation, Mean value theorems and applications, Linear approximation, Power Series, Taylor's Series, Maxima and Minima. Orthogonal Trajectories.

UNIT-3: Integration: Definite Integrals and their properties, Riemann Sums, Riemann Integral, The fundamental theorem of Integral Calculus and Improper Integrals. Application of Integration:

UNIT-4: Introduction and Motivation to Differential Equations, Order and Degree of Ordinary differential equations (ODEs); First Order ODEs: Geometrical Interpretation of Solution, Picard's Theorem for IVP and Picard's iteration method, Exact equations, integrating factor, linear equations, Euler's Method, Modified Euler's Method; Linear ODE of second and higher order with constant and variable coefficients, non-homogenous ODEs, power series solutions to second order ODEs.

UNIT-5: Sturm-comparison theorem, Sturm-Liouville BVP, Orthogonal Functions, Fourier Series and integrals, Laplace Transform and its application to ODEs.

Text and Reference Books:

1. George F. Simmons. Differential Equations With Applications and Historical Notes. 2nd ed. International Series in Pure & Applied Mathematics. McGraw-Hill Higher Education, 1991. ISBN: 978-0070575400.
2. George B. Thomas, Ross L. Finney, Calculus and analytical geometry, 9th edition, Pearson Publications, India, 2006.
3. Bartle and Sherbert, Introduction to real Analysis, Fourth Edition, John Wiley & Sons, 2011.

Assessment Method: Two MID-SEM + END-SEM + Project + Tutorial + Quizzes.

- 30% Marks based on Mid-Sem Examinations.
- 50% Marks based on End-Sem Examination.
- 20% Marks based on Quizzes and Project.

Course Outcomes: This course aims at building an advance understanding of calculus in single variable and ordinary differential equation. After successful completion of the course the student will have the ability to

- Formulate mathematical model of real-world situations by applying abstract concepts to complex problems and solving the model either analytically or by adopting numerical methods. (**P1, P2, P3, P4, P12**).

P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12
X	X	X	X								X

Lecture Schedule:

Sl. No.	Description		No. of Lectures
1	Real Number Systems		8
	1.1	Properties of Real Numbers	
	1.2	Sequence and Series	
	1.3	Limit, Continuity and Differentiability of Functions of Single Variable	
2	Differential Calculus		7
	2.1	Implicit Differentiation	
	2.2	Mean Value Theorems and Applications	
	2.3	Linear Approximations	
	2.4	Power Series and Taylor Series	
	2.5	Maxima and Minima, Orthogonal Trajectories	
3	Integral Calculus		7
	3.1	Definite Integrals and Their Properties	
	3.2	Riemann Integral	
	3.3	Improper Integrals	
	3.4	Application of Integration	
4	Ordinary Differential Equations		12
	4.1	Order and Degree of ODEs	
	4.2	Picard's Theorem for Initial Value Problems (IVPs)	
	4.3	Methods for Solving First Order Differential Equations	
	4.4	Euler's Methods for IVPs	
	4.5	Higher Order Linear ODEs with constant coefficients	
	4.6	Power Series Solutions for 2 nd Order ODEs	
	4.7	Methods of Variation of Parameter	
	4.8	Methods of Undetermined Coefficients	
5	Important Results and Laplace Transform		8
	5.1	Sturm Comparison Theorem	
	5.2	Sturm – Liouville Boundary Value Problem	
	5.3	Fourier Series and Integrals	
	5.4	Laplace Transform: Existence and its applications	